



Sydney Technical High School

Year 12 2 Unit Mathematics

HSC Assessment Task 2 - March 2003

Name: _____

Class: _____

Time Allowed: 70 minutes

Instructions:

1. Answer questions on paper provided
2. Begin each question on a fresh page.
3. Marks may be deducted for careless or untidy work
4. Show all working
5. Marks for each question are indicated next to the question. These marks are a guide and may be adjusted slightly if necessary.
6. Approved calculators may be used.

Question	1	2	3	4	5
Mark					
Total					

Question 1 (12 marks)

- 5 (a) Find: i) $\int (4x^2 + 6) dx$
- ii) $\int \frac{1}{x^2} dx$
- iii) $\int \sqrt{2x+1} dx$
- iv) $\int_1^4 (x^2 + 4) dx$
- 4 (b) Find the equation of a curve for which $y'' = 6x - 4$ and when $x = 1, y = 12$ and $y' = 7$.
- 3 (c) If $f(x) = \sqrt{2x^2 + 4}$, find $f''(x)$.

Question 2 (12 marks) (Begin a new page)

- 2 (a) Use calculus to find the values of x for which the curve $y = 4 + x - x^2$ is decreasing.
- 4 (b) Evaluate $\sum_{n=1}^{40} 3n - 1$
- 3 (c) For a certain function, $f'(x) = \frac{(x-2)(x-4)^2}{\sqrt{x(x+2)^3}}$.
- i) Give a reason why the function has turning points when $x = 2$ and $x = 4$.
- ii) Determine the nature of the turning point at $x = 2$.
- 3 (d) Julie is building a huge deck using 151 timber planks which decrease uniformly in length from 2500 mm to 400 mm so that the lengths of the planks form an arithmetic sequence. Find
- i) the difference in length between adjacent planks.
- ii) the total length (in metres) of planks needed.

Question 3 (10 marks) (Begin a new page)

- 4 (a) Find the values of x for which the curve $y = 4x^3 - 12x^2 + 2$ is
- i) concave up
 - ii) concave down
- 4 (b) The ground floor of a twenty story office block will cost \$200 000 to construct. The next floor will cost \$230 000, and the next, \$264 500. The cost of the remaining 17 floors will follow the same pattern. Find the total cost of building the twenty floors.
- 2 (c) The point $(1, 6)$ lies on the curve $y = f(x)$. If $f''(x) = 12(x - 1)^2$, determine whether or not $(1, 6)$ is a point of inflexion.

Question 4 (11 marks) (Begin a new page)

For the curve $y = x^3 - 6x^2 + 9x$, $-1 \leq x \leq 4$.

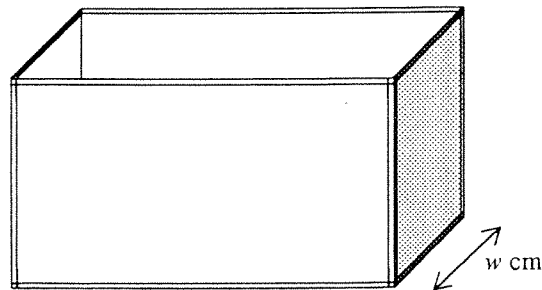
- 2 (a) Find y' and y'' .
- 4 (b) Find the coordinates of any stationary points and determine their nature.
- 2 (c) Find where the curve touches the x axis.
- 2 (d) Sketch the curve in the given domain showing all features determined above.
- 1 (e) Find the minimum value of the curve in the given domain.

Question 5 (10 marks) (Begin a new page)

- 4 (a) On a half page number plane diagram, sketch a possible curve for $y = f(x)$ which satisfies the conditions given in the following table:

x	0	1	2	3
$f(x)$		0	-1	
$f'(x)$	-1	0	-1	0
$f''(x)$		0	0	1

- 6 (b) An open cardboard box is twice as long as it is wide. The volume is 24cm^3 and all edges of the box are to be taped.



- Show that the length of tape needed is $12w + \frac{48}{w^2}$ where w is the width (in cm) of the box.
- Find the dimensions of the box which will give a minimum amount of tape.

End of Test

2003 24 HSC ASSESSMENT #2 SOLUTIONS.

Q1 a) i) $\int (4x^2+6)dx$
 $= \frac{4x^3}{3} + 6x + c$

ii) $\int \frac{1}{x^2} dx = \int x^{-2} dx$
 $= -x^{-1} + c$
 or
 $\frac{-1}{x} + c$

iii) $\int \sqrt{2x+1} dx = \int (2x+1)^{\frac{1}{2}} dx$
 $= \frac{(2x+1)^{\frac{3}{2}}}{\frac{3}{2} \times 2} + c$
 $= \frac{(2x+1)^{\frac{3}{2}}}{3} + c$

iv) $\int_1^4 (x^2+4) dx = \left[\frac{x^3}{3} + 4x \right]_1^4$
 $= \left(\frac{64}{3} + 16 \right) - \left(\frac{1}{3} + 4 \right)$
 $= 33$

b) $y'' = 6x - 4$

$\therefore y' = 3x^2 - 4x + c$

but $y' = 7, x = 1$

$\therefore 7 = 3 - 4 + c \Rightarrow c = 8$

$\therefore y' = 3x^2 - 4x + 8$

$y = x^3 - 2x^2 + 8x + k$

but $y = 12, x = 1$

$\therefore 12 = 1 - 2 + 8 + k \Rightarrow k = 5$

$\therefore y = x^3 - 2x^2 + 8x + 5$ is equation of the curve.

c) $y = (2x^2+4)^{\frac{1}{2}}$

$y' = \frac{1}{2}(2x^2+4)^{-\frac{1}{2}} \times 4x$

$= 2x(2x^2+4)^{-\frac{1}{2}}$

$y'' = -\frac{1}{2} \times 2x(2x^2+4)^{-\frac{3}{2}} \times 4x$
 $+ 2(2x^2+4)^{-\frac{1}{2}}$

$= 2(2x^2+4)^{-\frac{1}{2}} - 4x^2(2x^2+4)^{-\frac{3}{2}}$

Q2(a) $y = 4 + x - x^2$

$y' = 1 - 2x$

decreasing when $1 - 2x < 0$

ie $2x > 1$

$x > \frac{1}{2}$

(b) $\sum_{n=1}^{40} 3n - 1$

$= 2 + 5 + 8 + \dots + 119$

$= \frac{40}{2}(2 + 119)$

$= 2420$

(c) i) $f'(2) = f'(4) = 0$

ii) when $x < 2, f'(x) < 0$

" $x > 2, f'(x) > 0$

$\therefore$ local minimum when $x = 2$.

(d) i) $a = 400, T_{151} = 2500$

$\therefore 400 + 150d = 2500$

$\therefore d = 14$

ii) $S_{151} = \frac{151}{2}(400 + 2500)$

$= 218950 \text{ mm}$

$= 218.95 \text{ m}$

Q3(a) $y = 4x^3 - 12x^2 + 2$

$y' = 12x^2 - 24x$

$y'' = 24x - 24$

i) concave up when $24x - 24 > 0$
 $\Rightarrow x > 1$

ii) concave down when $x < 1$

(b) G.P. where $a = 200000$

$r = 1.15, n = 20$

$S_{20} = 200000 \frac{(1.15^{20} - 1)}{0.15}$

$= 20,488,717$

Q3(c) $f''(x) = 12(x-1)^2$
 $(1,6)$ is NOT a pt of inflexion since $f''(x)$ is always positive (does not change sign).

Q4. $y = x^3 - 6x^2 + 9x$,
 $-1 \leq x \leq 4$.

a) $y' = 3x^2 - 12x + 9$

$y'' = 6x - 12$

b) St. points occur when $y' = 0$

ie $3x^2 - 12x + 9 = 0$

$x^2 - 4x + 3 = 0$

$(x-3)(x-1) = 0$

$\therefore x = 1, 3$.

When $x=1$, $y=4$ and $y'' < 0$

$\therefore (1,4)$ is a local max.

When $x=3$, $y=0$ and $y'' > 0$

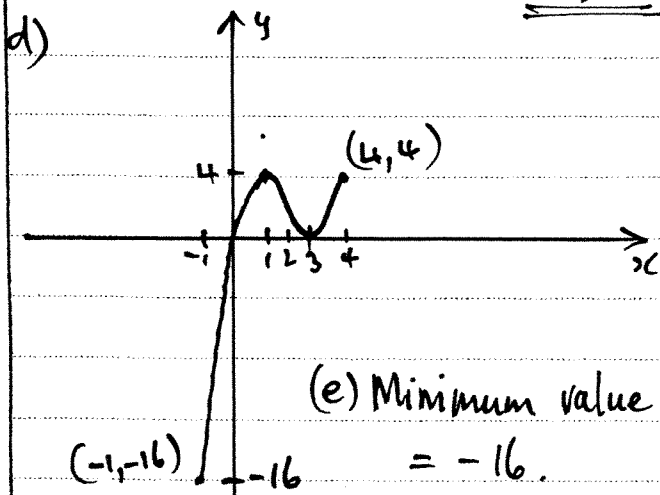
$\therefore (3,0)$ is a local min.

c) $y = 0 \Rightarrow x^3 - 6x^2 + 9x = 0$

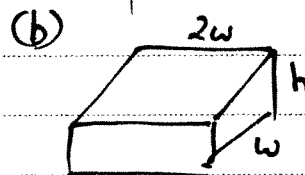
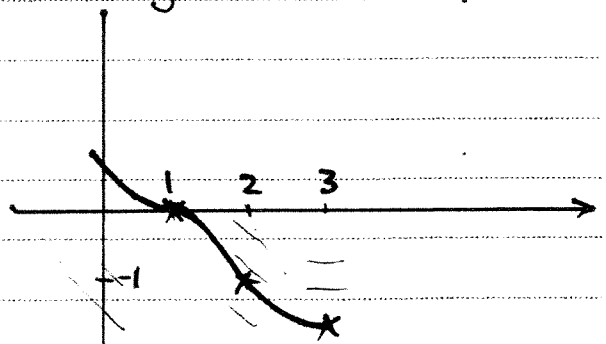
$x(x^2 - 6x + 9) = 0$

$x(x-3)^2 = 0$

$\therefore x = 0, 3$.



Q5.(a) Rough in with pencil first then go over with ink.



$V = 2w^2h = 24$

$\therefore h = \frac{12}{w^2}$ ✓

i) $L = 4 \times 2w + 4 \times w + 4 \times h$
 $= 12w + 4h$
 $= 12w + \frac{48}{w^2}$ ✓

ii) Min L occurs when $L' = 0$ and $L'' > 0$.

$L' = 12 - \frac{96}{w^3}$

$L'' = \frac{288}{w^4}$ which is $288w^{-4}$ always +ve.

When $L' = 0$, $\frac{96}{w^3} = 12$

ie $12w^3 = 96$

$w^3 = 8$

$\therefore w = 2$.

$\therefore$ Dimensions are $2 \times 4 \times 3$ cm.